SANSIKA KODITHUWAKKU

Software Engineer | Backend & Full-Stack | Python, SQL, PHP, React, Flutter

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EXECUTIVE SUMMARY

Software Engineer with over three years of experience building backend, full-stack and mobile systems in fintech. I launched a production mobile loan-origination product, automated bank payments, and cut reporting time for 1M+ monthly transactions from 4 hours to 15 minutes. I design secure REST APIs (JWT, RBAC, 2FA), use Redis caching and CI/CD, and work across Python, PHP, SQL, React and Flutter. I am completing an MSc in Computer Science at the University of Eastern Finland and have two publications in machine learning and retail analytics. Available part-time, up to 30 hours per week.

KEY FIGURES

- **9 systems** built at Abans Finance, from payments and lending to document management and reporting.
- **LKR 70 million** in sales through FinAb since Feb 2026; a 10-hour manual process now takes 20 minutes.
- **1M+ monthly transactions** reported on; report time cut from 4 hours to 15 minutes (93%).
- **1–2 business days to 4–5 minutes** for bank payments, including approvals.
- **98%+ payment success rate** on the Seylan Merchant Payment Gateway.
- **19 web applications** delivered at Cerfel; **2 publications** (IEEE ICAC 2024 and Propulsion Technology Journal 2023).

CORE SKILLS

Languages: Python, SQL, PHP, TypeScript, JavaScript, Java, Kotlin, C#, C++, C
Backend and APIs: FastAPI, Flask, Node.js, Express.js, NestJS, REST APIs, Microservices, WebSockets, Socket.IO, JWT, RBAC, 2FA
Databases: PostgreSQL, MySQL, MongoDB, Redis
Cloud and DevOps: Huawei Cloud, AWS (S3), Azure, Docker, GitHub Actions, Jenkins, CI/CD, Linux, Git
Frontend and Mobile: React, Redux, Material-UI, Tailwind CSS, HTML5, CSS3, Android (Java), Flutter, React Native
Testing and Practice: Unit, Integration, System and Acceptance (UAT) Testing, System Design, Agile/Scrum
AI-Assisted Development: Cursor (primary IDE), Claude (test automation and QA), ChatGPT (research and test design)

WORK EXPERIENCE

Senior Executive – Software Engineer (Full-Stack), Abans Finance PLC

Oct 2023 – Aug 2026

Sri Lanka | Promoted from Junior Executive (Grade A) to Senior Executive (Grade C) | Fintech payments, lending and internal systems

Fintech payments and mobile financing

FinAb – Mobile Financing Product

Live in production · Feb 2026 – present

Area: Fintech lending · Mobile · WorkHub24 platform · Owner: end to end

Context. Sales representatives needed a faster way to capture customer details, check credit and originate loans; the manual process took about 10 hours. FinAb was built on WorkHub24, a process-automation and digitisation platform widely used in the Sri Lankan industry.

My contribution

- Built the full application flow, document upload, status tracking, customer notifications through APIs and an operations dashboard.
- Wrote the custom APIs in PHP with authentication-header security and embedded them in the WorkHub24 platform.
- Automated the CRIB credit check, loan granting and loan origination inside the Abans environment.
- Prepared the product for vulnerability assessment and penetration testing (VAPT): hardened the server configuration (.htaccess) and ran a full check of the live APIs, then resolved findings before release approval.

Impact

- About LKR 70 million in sales since launch (Feb 2026).
- A manual process of about 10 hours now takes about 20 minutes.
- Released to production after VAPT.

Tech stack: PHP, REST APIs, SQL (company database), WorkHub24, CI/CD for API changes

WorkHub24 ↔ Seylan Bank Payment Automation (CEFT/SWIFT)

Production

Area: Treasury and payments · Secured API architecture · Owner: API and payment flow

Context. Utility and customer payments were approved inside WorkHub24 and then processed by hand, taking one to two business days with paper documentation.

My contribution

- Built platform-specific authorised APIs and secured APIs that connect the approval workflow to the bank.

- Implemented the full payment flow: the payment is made from the company account and the general-ledger (GL) record is written to the table.
- Stored every payment in a separate database for traceability, and generated a reference document for emergency use.

Impact

- Payment processing cut from 1–2 business days to 4–5 minutes, including approvals.
- All manual documentation removed.

Tech stack: Secured REST APIs, SQL, WorkHub24, Seylan Bank integration (CEFT/SWIFT)

Seylan Merchant Payment Gateway Integration

Production

Area: Customer payments · Frontend and backend

Context. Customers needed to pay instalments online, with payments reconciled in real time instead of by hand.

My contribution

- Integrated the gateway end to end: frontend, backend, gateway callbacks (webhooks) and transaction status handling.
- Implemented refunds and voids, and real-time payment reconciliation.
- Secured the flow with hosted checkout, tokenisation, idempotent processing, retries, an audit trail and careful handling of sensitive data.

Impact

- Payment success rate above 98%.
- Manual reconciliation reduced; customers paid on time more often.
- Transaction volume grew, and customer calls praised the time savings.

Tech stack: Payment gateway APIs, callbacks/webhooks, REST APIs, SQL, frontend integration

Internal mobile applications

Mobile App for Marketing and Recovery Officers

User-testing stage

Area: Internal field operations · Android (Java), Flutter · Version 1 and Version 2

Context. Marketing and recovery officers needed one app to find leads, initiate and track purchase orders, follow up customers, record field work and report. Managers also needed proof that field visits really happened.

My contribution

- Version 1: built the Android (Java) app with a PHP backend.
- Version 2: rebuilt it as a cross-platform Flutter app with Python and C# backend services; kept feature parity, redesigned state management and added security changes to the APIs.
- Added JWT authentication and Microsoft Authenticator sign-in; tested and optimised performance.
- Integrated Google Maps APIs with geo-tagging to track the actual distance travelled by each officer.

Impact

- More than 2x time saved compared with Version 1.
- Fewer false field-visit claims, lower fuel cost and a lower error rate.
- Better lead and purchase-order tracking.

Tech stack: Flutter, Android (Java), PHP, Python, C#, JWT, Microsoft Authenticator, Google Maps APIs

Backend systems, data and reporting

Document Management System (DMS)

Internal system

Area: Document control and compliance · Python backend, React frontend

Context. Loans could not proceed until original documents arrived, which took two business days. The DMS lets verified digital copies be used instead, with controlled access.

My contribution

- Built the backend with FastAPI and the frontend with React; stored files in an S3 bucket; automated build and deployment with Docker and CI/CD.
- Implemented two-factor authentication (pyotp), role-based access for six roles (Super Admin, Admin, Manager, Head Office Officer, Department Head, User) and watermarking (Pillow).
- Implemented live data synchronisation and audit logging with WebSockets.
- Indexed and hashed documents to speed up loading; supports PDF, Word and other document types.

Impact

- Removed the two-business-day wait for originals before loan processing.
- Faster document loading and a full audit trail for compliance.

Tech stack: Python, FastAPI, React, SQL, AWS S3, Docker, CI/CD, pyotp, Pillow, WebSockets

CRIB Credit Bureau Customer Viewer

Production

Area: Credit checks · Role: project lead and manager of junior developers

Context. More than 100 employees needed fast, secure access to credit bureau data, generating over 10,000 API calls a day.

My contribution

- Led the project, guided junior developers and managed delivery.

- Integrated the CRIB API with two-factor authentication; designed a 30-day automatic re-check policy as the cache-invalidation rule.
- Added Redis caching and query optimisation; built role-specific dashboards.

Impact

- Latency reduced by 35%.
- 100+ employees served and 10,000+ daily API calls handled.

Tech stack: PHP (backend and frontend), SQL, Redis, CRIB API, 2FA

High-Volume Reporting System

Production

Area: Finance reporting · 1M+ monthly transactions

Context. Reports over more than one million transactions a month took about four hours and slowed down other requests.

My contribution

- Built the system with dynamic schema generation so report columns and filters adapt to the data.
- Added a caching layer to cut waiting time and handle more concurrent requests; secured access with 2FA.

Impact

- Report generation cut from 4 hours to 15 minutes (93%).
- Scaled to 1M+ monthly transactions with shorter waiting times.

Tech stack: React, PHP, SQL, caching layer, 2FA

Purchase Order (PO) Management System

Production

Area: Procurement workflow · Marketing and finance teams

Context. PO approvals were manual and slow, and marketing needed to track PO progress.

My contribution

- Built role-based approval workflows, real-time notifications and automated PDF generation.
- Gathered requirements with the Marketing department and resolved ambiguity in the PO tracking process.

Impact

- Approval delays reduced by 40%.

IT Infrastructure Task Manager

Production

Area: IT operations · REST APIs and Slack integration

Context. The IT team tracked tasks in Excel, which was slow and hard to share.

My contribution

- Built REST APIs, replaced the Excel workflow and added Slack integration; optimised MySQL queries.

Impact

- Team productivity improved by 30%.
- MySQL query times cut from 2 minutes to 15 seconds.

Tech stack: REST APIs, MySQL, Slack integration

Security, architecture and collaboration

Security, Architecture and Collaboration

Across all systems

Area: API security · Microservices · Testing

My contribution

- Implemented JWT authentication, RBAC and API security aligned with Central Bank API security and governance requirements and the Personal Data Protection law.
- Designed fintech services as microservices and tested them at unit, integration, system and user-acceptance (UAT) levels.
- Automated builds, tests and server deployments with GitHub Actions; hosted backends and frontends on Huawei Cloud (storage and server configuration).
- Gathered requirements with finance, operations and marketing teams and provided application support alongside development.

Tech stack: JWT, RBAC, GitHub Actions, Docker, Huawei Cloud, unit/integration/system/UAT testing

Software Engineer Intern, Cerfel (Pvt) Ltd.

2023 (6 months)

Sri Lanka | Web development for tourism clients

Cerfel – 19 Full-Stack Web Applications for Tourism Clients

2023 · 6 months

Area: Client web development · Booking, user data and analytics

Context. Tourism clients needed custom web applications with admin panels to manage bookings, users and analytics.

My contribution

- Delivered 19 full-stack applications with custom admin panels for booking management, user data and analytics.
- Refactored code and added React Query and caching; documented workflows and API specifications.

Impact

- Application performance improved by 40%.
- New-developer onboarding time cut by 25%.

Tech stack: React, TypeScript, Redux, React Query, Node.js, PHP, PostgreSQL

PROJECTS AND RESEARCH

Fake News Detection – Mixture of Experts (Research and Platform) Published · IEEE ICAC 2024

Area: Machine learning research and software platform · Thesis

Context. Misinformation detection improves when sentiment and sarcasm are modelled explicitly. The thesis used them as core expert models in a Mixture-of-Experts design.

My contribution

- Wrote the paper, and developed and tested the models.
- Built the platform frontend and backend with a browser extension and a mobile app; deployed as Docker microservices with a Jenkins CI/CD pipeline on cloud infrastructure.

Impact

- Published: DOI 10.1109/ICAC64487.2024.10851056.

Tech stack: Python, NestJS, React, Redux Toolkit, Docker, Jenkins, browser extension, mobile app

Collaboard – Real-Time Collaborative Whiteboard Personal project

Area: Real-time collaboration

My contribution

- Built a multi-user drawing application with room-based real-time synchronisation, customisable brushes and boards stored in MongoDB.

Impact

- Sub-100 ms latency for drawing actions.

Tech stack: React, Socket.IO, MongoDB

e-Sarasaviya – Learning Management System Web application

Area: Education technology

My contribution

- Built a learning management system with a class scheduler for organising classes.

Tech stack: Web application

University E-Commerce Platform University project

Area: E-commerce

My contribution

- Developed and maintained an e-commerce application with a full backend in C# and a TypeScript frontend.

Tech stack: C#, TypeScript, C++/C (coursework)

OVERALL TECHNICAL CONTRIBUTION

Most of my professional work was in fintech at Abans Finance, where I owned systems from requirements to production: payments, lending, document control, reporting and field mobile apps. I worked across backend (Python, PHP, C#), frontend (React) and mobile (Android, Flutter), and across the layers of each system: APIs, security, caching, data and deployment. The work shows steady growth from building features to owning products, leading a small team on the CRIB viewer and delivering measurable performance and time savings.

PUBLICATIONS

- **Improving Fake News Detection with a Mixture of Experts: Leveraging Sentiment Analysis and Sarcasm Detection as Core Expert Models.** IEEE, ICAC 2024. DOI: 10.1109/ICAC64487.2024.10851056
- **User Based Sale Optimization for Fashion Retail.** Propulsion Technology Journal, Vol. 44, No. 5 (2023).

EDUCATION

MSc in Computer Science (International Master's Degree Programme in Information Technology) In progress

University of Eastern Finland, Joensuu

BSc (Hons) in IT, Specialising in Software Engineering – Second Class Upper Division 2021 – 2025

SLIIT, Malabe, Sri Lanka

CERTIFICATIONS

- Microsoft Applied Skills: Secure AI solutions in the cloud
- Microsoft Applied Skills: Get started with cloud security and monitoring tasks

AWARDS, LANGUAGES AND REFERENCES

Awards and leadership: Dean's List (4th year, 2nd semester 2024) | Rotaract International Service Director | Board Member of the Month (twice)

Languages: English – Expert (C1; IELTS Overall 7.0) | Finnish – Basic | **References:** available on request.